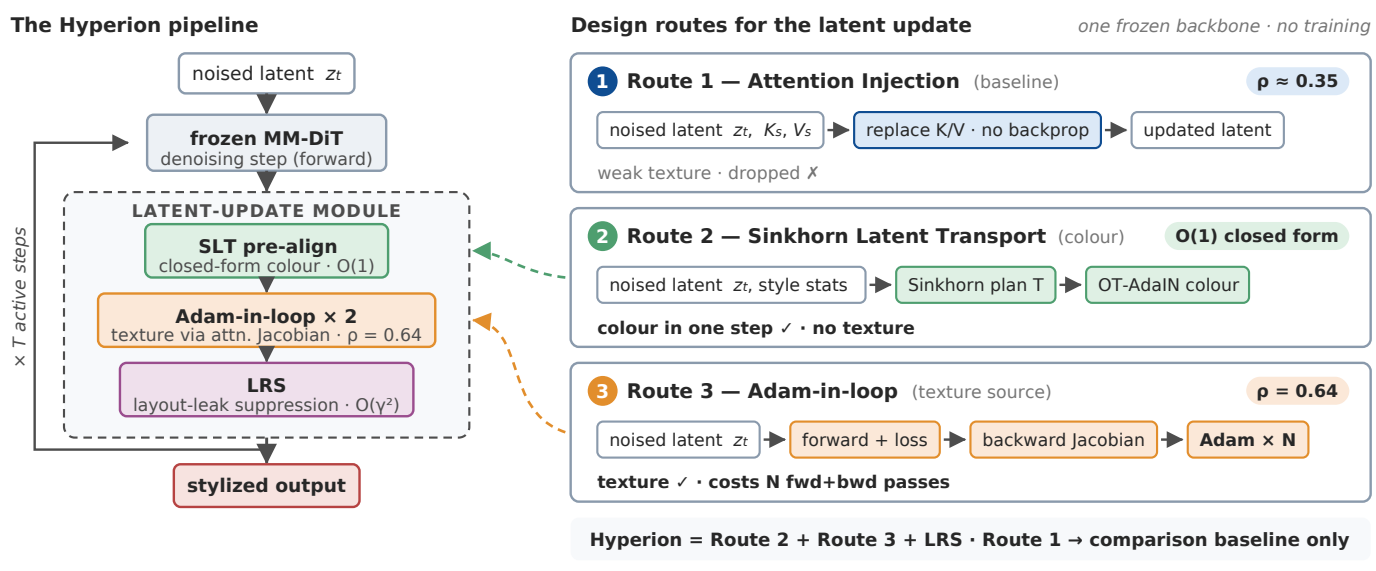
The Hyperion pipeline
noised latent z_t
frozen MM-DiT
denoising step (forward)
LATENT-UPDATE MODULE
SLT pre-align
closed-form colour · O(1)
Adam-in-loop × 2
texture via attn. Jacobian · ρ = 0.64
LRS
layout-leak suppression · O(γ²)
× T active steps
stylized output
Design routes for the latent update
one frozen backbone · no training
1 Route 1 — Attention Injection (baseline)
ρ ≈ 0.35
noised latent z_t, K_s, V_s
replace K/V · no backprop
updated latent
weak texture · dropped ✗
2 Route 2 — Sinkhorn Latent Transport (colour)
O(1) closed form
noised latent z_t, style stats
Sinkhorn plan T
OT-AdaIN colour
colour in one step ✓ · no texture
3 Route 3 — Adam-in-loop (texture source)
ρ = 0.64
noised latent z_t
forward + loss
backward Jacobian
Adam × N
texture ✓ · costs N fwd+bwd passes
Hyperion = Route 2 + Route 3 + LRS · Route 1 → comparison baseline only